

AI Product Management Certification

by Product Faculty — Rohan Varma (OpenAI) & Henry Shi (Anthropic)

PART 1 — Modules 1 to 5

Expanded Lecture Notes with Real-World Case Studies

How to Use These Notes

These notes are written in lecture style — as if Rohan and Henry are teaching you live. Every framework is explained with real case studies, worked examples, and actionable templates. Use these notes to teach your students, build your portfolio, and ace AI PM interviews.

MODULE 1: Identify High-Leverage AI Opportunities Using the 4D Method

Instructors: Rohan Varma (OpenAI) & Henry Shi (Anthropic)

Lesson 1.1 — Start With User Pain, Not Technology

Rohan speaking:

The #1 mistake AI PMs make is starting with the technology. "We have access to GPT-4, what should we build?" This is backwards. The right question is: "What is the most painful thing our users do today, and can AI solve it better than anything else?" If you start with the technology, you end up with a solution looking for a problem. If you start with user pain, you end up with a product people actually need.

The AI Opportunity Stack

Layer	Focus	Key Question
Layer 1: User Pain	Start here — always	What is the most frequent, high-severity problem users face?
Layer 2: Solution Gap	Why existing solutions fail	Why do current solutions fail? What is the workaround?
Layer 3: AI Suitability	Is AI the right tool?	Is this a pattern recognition problem? Is there enough data? Is the cost of being wrong acceptable?
Layer 4: Business Value	Measurable impact	If AI solves this, what is the measurable business impact?

Case Study: Notion AI

REAL CASE STUDY — Notion AI:

Layer 1 — Pain: Knowledge workers spend 2-3 hours/day writing first drafts. Pain Score: 88/100.

Layer 2 — Gap: Templates help with structure but not content generation. Users still stare at blank pages.

Layer 3 — AI Suitability: Writing is a language task — perfect for LLMs. Cost of wrong answer: low (user edits).

Layer 4 — Business Value: Saves 1 hour/day per user = \$25/user/day in productivity. Massive retention driver.

RESULT: Notion AI reached 1M users in 3 months. Started with pain, not technology.

Anti-Pattern: Technology-First

ANTI-PATTERN — Technology-First Thinking:

A major enterprise company built an "AI-powered dashboard" because their CTO was excited about GPT-4.

6 months of work. Adoption: 3% of users. Why? Nobody asked users what their actual pain was.

Post-launch research: users needed faster data entry, not smarter dashboards. \$2M and 6 months wasted.

Lesson 1.2 — The 4D Method

Henry speaking:

The 4D Method is your systematic framework for finding where AI creates real value. The PMs who consistently identified the best opportunities all followed a similar pattern — they mapped workflows, identified pain, tested AI suitability, and then chose the right approach.

D	Step	Key Question	Output
Discover	Map all user workflows	Where do users spend the most time? Where do they make the most errors?	Workflow map with time/error annotations
Define	Identify highest-pain workflow	Which workflow, if improved, would have the biggest impact on user success?	Prioritised pain list with scores
Determine	Apply AI Suitability Test	Is AI the right solution? Or is this a process/UX problem?	AI Suitability Score (0-100)
Decide	Choose the AI approach	RAG, Agent, Classifier, or Generator — which fits the problem?	AI Approach Decision with rationale

Case Study: Legal Tech 4D Method

REAL CASE STUDY — Legal Tech Company:

DISCOVER: Mapped 8 workflows for corporate lawyers. Contract review scored highest on pain.

DEFINE: Lawyers spend 4-6 hours reviewing a single contract. Error rate: 12%. Workaround: expensive junior associates.

DETERMINE: AI Suitability Score: 82/100. Pattern recognition + language understanding + data available.

DECIDE: RAG + Classifier. RAG for legal precedents. Classifier to flag high-risk clauses. NOT a full agent — too high-stakes.

RESULT: Review time: 4 hours → 45 minutes. Error rate: 12% → 3%.

Lesson 1.3 — The 5-Question Framework to Quantify Pain

Rohan speaking:

You need to quantify pain so you can prioritise objectively and build a business case. The 5-Question Framework gives you a Pain Score from 0 to 100.

Question	What It Measures	Score 5	Score 3	Score 1
Frequency	How often does this pain occur?	Daily	Weekly	Monthly
Severity	How bad is it when it happens?	Blocks work completely	Slows work significantly	Minor inconvenience
Population	How many users experience this?	>80% of users	40-80% of users	<40% of users
Workaround	How painful is the current workaround?	No workaround exists	Bad/expensive workaround	Acceptable workaround
Unlocks	What does solving this enable?	New revenue stream	Retention improvement	Efficiency gain only

Pain Score Formula + Worked Example

Pain Score = (Frequency + Severity + Population + Workaround + Unlocks) / 25 x 100

Score > 70: Strong AI opportunity. Build the business case.

Score 40-70: Investigate further. Run more user research.
Score < 40: Probably not worth it. Find a higher-pain problem.

WORKED EXAMPLE — Salesforce CRM Auto-Update:

Problem: Sales reps spend 45 min/day manually updating CRM records after calls.

Frequency: Daily = 5 | Severity: Slows work + data quality issues = 4 | Population: 90% of reps = 5

Workaround: Manual entry (painful, error-prone) = 4 | Unlocks: Better forecasting = revenue impact = 4

Pain Score = $(5+4+5+4+4) / 25 \times 100 = 88/100$

DECISION: Strong AI opportunity. This became Salesforce Einstein Call Coaching — adopted by 500K+ reps.

Lesson 1.4 — The AI PRD: What Is Different

Henry speaking:

A standard PRD has problem statement, user stories, success metrics, and out of scope. An AI PRD has all of that PLUS 8 additional sections that most PMs skip — and that is why most AI products fail.

AI PRD Section	What to Include	Why It Matters
AI Approach	RAG / Agent / Classifier / Generator — and why	Aligns engineering on architecture from day one
Data Requirements	What data the model needs, quality bar, labelling needs	Data problems kill AI projects. Define this early.
Model Selection Rationale	Which model tier and why (cost vs quality trade-off)	Wrong model = 10x cost or 30% worse quality
Prompt Strategy	High-level prompt architecture and iteration plan	Prompts are product decisions, not engineering details
Evaluation Plan	Golden Dataset size, judge rubric, key metrics	Without evals, you cannot know if you improved
Guardrails Specification	Input guardrails, output guardrails, human oversight	Safety is a product requirement, not an afterthought
Cost Model	Tokens x price x volume = monthly cost estimate	AI features can bankrupt a product if unplanned
AI Risk Register	Top 5 risks with probability, severity, and mitigation	Stakeholders need to see you have thought about failure

Anti-Pattern: No AI PRD

CASE STUDY — The \$45K Surprise:

A fintech startup built an AI expense categorisation feature. No AI PRD. No cost model.

Month 1: Feature launched. Users loved it.

Month 2: 50,000 users. Monthly AI cost: \$45,000. Budget: \$5,000.

Month 3: Feature shut down. Users furious. Trust destroyed.

The cost model would have taken 2 hours to build:

50K users x 20 queries/day x \$0.045/query = \$45K/month

They would have used a cheaper model, built caching, or priced it differently.

The AI PRD is not bureaucracy. It is survival.

Module 1 Lab Assignment

Lab Assignment

1. Build an AI Opportunity Assessment using the 4D Method for a product you know well.
Discover: Map at least 5 user workflows with time and error annotations.
Define: Score each workflow on the 5-Question Framework. Identify the top pain.
Determine: Run the AI Suitability Test. Score 0-100.
Decide: Choose RAG, Agent, Classifier, or Generator. Write 3 sentences justifying your choice.
2. Write a 1-page AI PRD with all 8 AI-specific sections.
Include a cost model at 10K, 100K, and 1M monthly queries.
Include a risk register with top 3 risks, probability x severity, and mitigation.

PORTFOLIO TIP: This becomes Artifact 1 of your AI PM Portfolio.

MODULE 2: Design Trustworthy AI Experiences That Drive Adoption

Lesson 2.1 — The #1 UX Mistake in AI Products

Rohan speaking:

The #1 UX mistake is adding a chat interface and calling it an AI product. Chat is one interaction pattern. It is not always the right one. Users do not want to talk to AI. They want to get their job done. AI should be invisible infrastructure that makes their existing workflow faster and better.

Pattern	Best For	Real Example	When NOT to Use
Chat / Conversational	Open-ended exploration, Q&A, support	ChatGPT, Intercom AI support bot	When the task is structured and repetitive
Inline Suggestion	Augmenting existing workflows	GitHub Copilot, Gmail Smart Compose	When suggestions would be distracting
Proactive Nudge	Surfacing insights users did not ask for	Slack channel summaries, CRM next-best-action	When users are in deep focus mode
Automated Action	Repetitive, low-stakes tasks	Auto-categorisation, auto-tagging in Notion	When the cost of error is high
Structured Output	Generating documents, reports, summaries	Notion AI page generator, meeting notes	When output needs heavy customisation

Case Study: Interaction Pattern Determines Adoption

CASE STUDY — GitHub Copilot vs. Chat-First Competitor:

GitHub Copilot chose INLINE SUGGESTION. Code appears directly in the editor as the developer types.

A competitor built a CHAT interface — developers had to describe what they wanted, copy the code, paste it.

Result: Copilot adoption: 1.5M paid users in Year 1. Competitor: shut down in 18 months.

The difference was NOT model quality. Both used GPT-4. The difference was interaction pattern.

Copilot fit into the developer workflow. The competitor asked developers to change their workflow.

Lesson 2.2 — The 3P Framework for AI UX

Henry speaking:

For every AI feature, evaluate it on three dimensions before you build anything. If a feature fails on any one of these, it will not get adopted — no matter how good the AI is.

P	Dimension	Key Question	Red Flag
P1	Prioritisation	Is this AI feature solving the highest-priority user pain? Or is it a nice-to-have?	Pain Score < 50
P2	Placement	Is the AI feature placed where users are already working? Or does it require them to go somewhere new?	Requires navigation away from primary workflow
P3	Prominence	Is the AI feature visible enough to be discovered? Or is it buried in a settings	Feature discovery rate < 20% in first week

		menu?	
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Case Study: Grammarly 3P Done Right

CASE STUDY — Grammarly 3P Analysis:
P1 Prioritisation: Writing errors are a daily, high-severity pain for professionals. Pain Score: 85/100. PASS.
P2 Placement: Grammarly integrates directly into Gmail, Google Docs, LinkedIn, Slack. Users never leave their workflow. PASS.
P3 Prominence: The Grammarly icon appears in every text field. Suggestions appear inline with a green underline. Impossible to miss. PASS.
RESULT: 30M daily active users. \$13B valuation.

Lesson 2.3 — Designing for Trust

Rohan speaking:

Trust is the #1 adoption driver for AI features. I have seen features with 90% accuracy fail because users did not trust them. I have seen features with 75% accuracy succeed because users trusted them. Trust is not about accuracy. Trust is about transparency, control, and recovery.

Trust Signal	How to Implement It	Real Example
Source Transparency	Show where the AI got its information	"Based on your last 3 meetings" — Otter.ai meeting summaries
Confidence Indicators	Show when the AI is uncertain	"I am not sure about this — you may want to verify" — Perplexity AI
Human Override	Always give users a way to edit, reject, or escalate	Gmail Smart Compose: press Tab to accept, keep typing to ignore
Explainability	Show the reasoning, not just the answer	"I recommended this because your last 3 purchases were similar" — Spotify
Error Recovery	Make it easy to report bad outputs and get a human alternative	"Was this helpful? Connect to human agent" — Intercom

Case Study: Perplexity AI Trust Model

CASE STUDY — Perplexity AI Trust Through Transparency:
Perplexity AI built trust by showing sources for every answer. Every claim is linked to a source URL. Users can verify any fact in one click. Confidence indicators show when the AI is uncertain.

Compare to early ChatGPT: no sources, no citations, confident even when wrong.
Perplexity grew to 10M DAU in 2024 largely because researchers and professionals trusted it more.

LESSON: In high-stakes domains (research, medical, legal, financial), source transparency is not optional. It is the product.

Lesson 2.4 — Designing AI Error States

Henry speaking:

AI fails differently from traditional software. Traditional software either works or crashes. AI degrades gracefully — or ungracefully, if you have not designed for it. There are 4 AI error states every PM must design for.

Error State	What Happens	Design Response	Real Example
Hallucination	AI generates confident but wrong information	Source citations, confidence scores, human review flag	Bing Chat citing fake legal cases in 2023
Refusal	AI declines to answer (over-cautious guardrails)	Clear explanation + alternative path (human handoff)	ChatGPT refusing to help with chemistry homework
Irrelevance	AI answers a different question than asked	Clarification prompt, "Did you mean..." pattern	Support bot answering billing question with product docs
Timeout/Failure	Model API is slow or unavailable	Graceful degradation, cached responses, human fallback	"AI temporarily unavailable — here are our top FAQs"

Anti-Pattern: Air Canada Chatbot

CASE STUDY — Air Canada Chatbot Hallucination (\$1M Lesson):

In 2024, Air Canada's AI chatbot told a customer he could book a bereavement fare and claim a refund later.

This was wrong. Air Canada's policy did not allow retroactive refunds.

The customer sued. Air Canada lost. The court ruled the chatbot's output was binding.

WHAT WENT WRONG: No hallucination guardrails. No source citations. No human review for high-stakes claims.

WHAT SHOULD HAVE BEEN BUILT:

- Source citations: "Based on our refund policy [link]"
- Confidence threshold: If confidence < 0.85, route to human agent
- High-stakes topic detection: Refund/legal queries always go to human review

LESSON: Error state design is legal and financial risk management.

Module 2 Lab Assignment

Lab Assignment

1. Map the interaction patterns for your AI feature using the matrix. Justify your choice.
2. Apply the 3P Framework — score your feature on Prioritisation, Placement, Prominence (1-10 each). Identify the weakest P and propose a design improvement.
3. Design the 5 trust signals for your specific feature. Write the exact UX copy for each.
4. Design all 4 error states with specific UX responses.
Write the exact error message copy. Define the threshold that triggers each error state.

MODULE 3: Systematically Optimize Outputs for Accuracy & Relevance

Lesson 3.1 — Why Output Quality Is a PM Problem

Henry speaking:

Most PMs treat output quality as an engineering problem. "The model will figure it out." This is wrong. Output quality is determined by three things: the model you choose, the data you provide, and the prompt you write. PMs own two of those three. If your AI outputs are bad, look in the mirror first.

Output Quality Formula

Output Quality = f(Model Choice x Data Quality x Prompt Quality)

PMs own: Data Requirements (in PRD) + Prompt Strategy (in PRD) + Model Selection Rationale (in PRD)

Engineers own: Model implementation + Infrastructure + Latency optimisation

Lesson 3.2 — The Prompt Engineering Hierarchy

Rohan speaking:

There is a hierarchy of prompt engineering techniques. Each level builds on the previous one. Most PMs stop at Level 1 or 2. The best AI PMs go to Level 4 or 5.

Level	Technique	When to Use	Quality Gain	Example
1	Zero-Shot	Simple, well-defined tasks	Baseline	"Summarise this document in 3 bullet points."
2	Few-Shot	Tasks with specific output format	+15-25%	Provide 3 examples of good summaries before asking
3	Chain-of-Thought	Reasoning tasks, multi-step problems	+20-35%	"Think step by step before answering."
4	Role + Context + Constraints	Domain-specific tasks	+25-40%	"You are a senior PM at a B2B SaaS company. Your job is to..."
5	RAG (Retrieval-Augmented)	Tasks requiring specific knowledge	+40-60%	Retrieve relevant docs, inject into context, then generate

Case Study: Intercom Support Bot

CASE STUDY — Intercom Support Bot: 47% Quality Improvement

BEFORE: Zero-shot prompt. "Answer the customer's question." Accuracy: 61%.

ITERATION 1: Added few-shot examples (5 good answers). Accuracy: 74%. (+13%)

ITERATION 2: Added chain-of-thought. "First, identify intent. Then check knowledge base. Then respond." Accuracy: 81%. (+7%)

ITERATION 3: Added role + context + constraints. "You are a senior Intercom support specialist. Always cite the help article. Never make up pricing." Accuracy: 87%. (+6%)

ITERATION 4: Added RAG — retrieve top 3 relevant help articles before generating. Accuracy: 91%. (+4%)

TOTAL: 61% to 91% (+47%) through systematic prompt engineering. No model change. No fine-tuning.

Lesson 3.3 — The System Prompt Architecture

Henry speaking:

A production system prompt has 6 components. Missing any one of them degrades quality. The ones that fail are almost always missing Component 4 (Constraints) or Component 5 (Output Format).

Component	What It Does	Example
1. ROLE	Define who the AI is in this specific context	"You are a senior product manager at a B2B SaaS company specialising in HR software."
2. TASK	Define exactly what the AI should do with the inputs	"Your job is to write user stories given a feature description and target user persona."
3. CONTEXT	Provide the relevant background information	"Here is our product documentation: [docs]. Here is our target user: [persona]."
4. CONSTRAINTS	Define what the AI must always do and must never do	"You must always include acceptance criteria. Never exceed 3 user stories. Never use technical jargon."
5. OUTPUT FORMAT	Define the exact structure of the response	"Format: As a [user type], I want [action] so that [benefit]. Acceptance criteria: [list]."
6. EXAMPLES (Few-Shot)	Provide 2-3 examples of ideal outputs	"Here is an example of a great user story: [example]."

Case Study: Jasper AI System Prompt

CASE STUDY — Jasper AI System Prompt Architecture:

Jasper AI (\$1.5B valuation) uses a 6-component system prompt for every content type.

For blog posts:

ROLE: "You are an expert content marketer with 10 years of experience in [industry]."

TASK: "Write a 1,500-word blog post on [topic] targeting [audience]."

CONTEXT: "Brand voice: [voice guide]. Target keyword: [keyword]. Competitor articles: [URLs]."

CONSTRAINTS: "Must include H2 headers every 300 words. Never use passive voice. Always end with a CTA."

OUTPUT FORMAT: "Title | Introduction (150 words) | 4 x H2 sections (300 words each) | Conclusion (150 words) | CTA"

EXAMPLES: 3 high-performing blog posts from the client's archive.

RESULT: Content quality scores 40% higher than zero-shot. Client retention rate: 89%.

Lesson 3.4 — RAG Architecture for PMs

Rohan speaking:

RAG — Retrieval-Augmented Generation — is the most important AI architecture pattern for PMs to understand. It solves the #1 problem with LLMs: they do not know your specific data. GPT-4 knows everything on the internet up to its training cutoff. It does not know your product documentation, your customer data, or your internal policies. RAG fixes this.

How RAG Works (Plain English):

- **Step 1 — User Query:** User asks: "What is your refund policy for annual subscriptions?"
- **Step 2 — Retrieval Engine:** Search your knowledge base for the top 3 most relevant chunks about refund policies.
- **Step 3 — Context Assembly:** Combine the retrieved chunks with the user query into a single prompt.
- **Step 4 — LLM Generation:** LLM generates an answer grounded in the retrieved context — not its training data.
- **Step 5 — Grounded Response:** User gets an accurate answer based on your actual policy, not a hallucinated one.

RAG Quality Lever	What It Controls	PM's Role
Chunking Strategy	How documents are split for retrieval	Define chunk size requirements based on use case (e.g. 512 tokens for support docs)
Retrieval Quality	How well the search finds relevant chunks	Define relevance criteria, test with 50 real user queries
Context Window	How much context is passed to the model	Balance quality vs. cost (more context = more tokens = more cost)
Grounding Instructions	How strictly the model sticks to retrieved context	Define faithfulness requirements: "Only answer from retrieved context. If not found, say so."

Case Study: RAG for HR Knowledge Base

CASE STUDY — RAG for Internal HR Knowledge Base:

PROBLEM: A 500-person company wants AI to answer questions about internal HR policies.

OPTION A — Vanilla ChatGPT: Does not know your specific policies. Hallucination rate: 35%.

OPTION B — RAG on HR Policy Documents: Retrieves relevant policy sections, generates grounded answers. Hallucination rate: 4%.

CHUNKING: HR policies chunked at 512 tokens with 50-token overlap.

RETRIEVAL: Semantic search (embeddings) + keyword search (hybrid). Top 3 chunks retrieved.

GROUNDING: "Only answer from the provided HR policy documents. If the answer is not in the documents, say: I could not find this in our HR policies — please contact HR directly."

RESULT: Employee satisfaction with HR queries: 34% to 81%. HR ticket volume: -42%.

Module 3 Lab Assignment

Lab Assignment

1. Write a production-quality system prompt using all 6 components for your AI feature.

2. Test your prompt at 3 levels of the hierarchy:

Level 1: Zero-shot. Score quality 1-10.

Level 3: Chain-of-thought. Score quality 1-10.

Level 4: Role + Context + Constraints. Score quality 1-10.

Document the quality difference at each level.

3. Design a RAG architecture for your AI feature:

Define your knowledge base (what documents/data).

Define your chunking strategy (chunk size, overlap).

Define your retrieval quality criteria.

Write your grounding instruction.

MODULE 4: Build Rigorous Evaluation Suites — Not Vibes

Lesson 4.1 — Why Evaluation Is the Most Important PM Skill in AI

Henry speaking:

Henry Shi, Anthropic

"If you can't measure it, you can't improve it. In AI, if you can't evaluate it, you're flying blind. Every prompt change, every model upgrade, every architecture decision — you need an eval suite to know if you made things better or worse. The PMs who build great AI products are not the ones with the best intuition. They are the ones with the best eval suites."

Most AI teams evaluate by vibes. They make a change, try a few examples, and say "yeah, that seems better." This is how you ship regressions. A rigorous eval suite is what separates amateur AI teams from professional ones.

Lesson 4.2 — The Golden Dataset

Rohan speaking:

The Golden Dataset is your ground truth. It is a curated set of inputs with known-good outputs that you use to measure your AI system's quality. Think of it as your test suite — but for AI. Every time you make a change to your AI system, you run it against the Golden Dataset and measure the delta.

Category	% of Dataset	What It Tests	Example
Happy Path	60%	Standard use cases your AI should handle well	Normal customer support questions with clear answers
Edge Cases	25%	Unusual inputs, ambiguous queries, boundary conditions	Questions in multiple languages, very long queries, typos
Adversarial	15%	Inputs designed to break your AI — jailbreaks, contradictions, trick questions	Prompt injection attempts, contradictory instructions, out-of-scope requests

Golden Dataset Guidelines + Case Study

Minimum Viable Golden Dataset: 30 examples (18 happy path, 7 edge cases, 5 adversarial)

Production Golden Dataset: 200+ examples

Enterprise Golden Dataset: 1,000+ examples with demographic diversity

CRITICAL: Your Golden Dataset must be built BEFORE you start building. Not after.

CASE STUDY — Duolingo:

Duolingo built a 500-example Golden Dataset before launching AI-generated content.

Before Golden Dataset, regression rate on model upgrades: 23%. After: 4%.

They caught a model upgrade that improved overall quality by 8% but degraded Spanish idiomatic expressions by 31%.

Without the eval suite, they would have shipped that regression.

Lesson 4.3 — The Evaluation Metrics Framework

Henry speaking:

Metric	What It Measures	How to Score	Target	Red Flag
Faithfulness	Is the output grounded in the provided context?	LLM-as-Judge (0-1)	>0.85	<0.70 = hallucination problem
Relevance	Does the output answer the actual question?	LLM-as-Judge (0-1)	>0.80	<0.65 = retrieval problem
Completeness	Does the output cover all required elements?	Rubric checklist	>0.75	<0.60 = prompt problem
Latency	How fast does the system respond?	p50, p95, p99 (seconds)	p95 < 3s	p95 > 5s = infra problem
Cost per Query	Token cost per user interaction	\$ per 1,000 queries	Defined by business case	>2x budget = model/prompt problem

Lesson 4.4 — LLM-as-Judge

Rohan speaking:

LLM-as-Judge is the technique of using a separate LLM to evaluate the outputs of your AI system. It scales qualitative evaluation that would otherwise require human reviewers. Research shows LLM-as-Judge achieves 85-90% agreement with human judges on well-defined rubrics.

LLM-as-Judge Prompt Template

LLM-AS-JUDGE PROMPT TEMPLATE:

You are an expert evaluator for an AI [use case] system.

Given the following:

- User Query: {query}
- Retrieved Context: {context}
- AI Response: {response}

Evaluate the response on:

1. FAITHFULNESS (0-1): Is every claim supported by the retrieved context? 1.0 = fully grounded, 0 = hallucinated.
2. RELEVANCE (0-1): Does the response directly answer the user's query? 1.0 = perfectly relevant, 0 = completely off-topic.
3. COMPLETENESS (0-1): Does the response cover all important aspects? 1.0 = comprehensive, 0 = missing key information.

Return your evaluation as JSON:

```
{
  "faithfulness": [score],
  "relevance": [score],
```

```
"completeness": [score],  
"overall": [average],  
"reasoning": "[1-2 sentence explanation]",  
"failure_mode": "[if overall < 0.7, describe the primary failure]"  
}
```

Case Study: Stripe LLM-as-Judge

CASE STUDY — Stripe LLM-as-Judge at Scale:

Stripe built an AI assistant for developer documentation. They needed to evaluate 10,000 queries/week.

HUMAN REVIEW: 10,000 queries x 2 min/review x \$25/hr = \$8,333/week. Not scalable.

LLM-AS-JUDGE: 10,000 queries x \$0.002/query (GPT-4o-mini as judge) = \$20/week.

VALIDATION: 200 queries through both human review and LLM-as-Judge. Agreement rate: 87%.

RESULT: Caught a prompt regression in Week 3 that dropped faithfulness from 0.89 to 0.71.

Fixed in 2 hours. Without LLM-as-Judge, this would have gone undetected for weeks.

Module 4 Lab Assignment

Lab Assignment

1. Build a 30-example Golden Dataset for your AI feature.
18 happy path, 7 edge cases, 5 adversarial.
For each example: input, expected output, evaluation rubric.
2. Write a custom LLM-as-Judge prompt for your use case.
Define your 3 evaluation dimensions with clear 0-1 scoring rubrics.
3. Run your Golden Dataset through your AI system and score every output.
Calculate your average Faithfulness, Relevance, and Completeness scores.
4. Identify your top 3 failure patterns and root causes.
Is it a retrieval problem? A prompt problem? A model problem?
5. Make one improvement and re-run — document the delta.
This becomes your Improvement Log (v1 to v2) for your portfolio.

MODULE 5: Architect Agentic AI Systems That Work

Lesson 5.1 — What Is an AI Agent?

Henry speaking:

An AI agent is an LLM that can take actions in the world — not just generate text. This is the fundamental shift from AI as a content generator to AI as an autonomous worker. When you ask ChatGPT a question, it generates text. When an AI agent gets a task, it plans, uses tools, observes results, and iterates until the task is done.

Layer	Function	Examples
Sense (Inputs)	Perceive inputs from the environment	User message, documents, API responses, tool outputs, database queries
Plan (Reasoning)	Decide what to do and in what order	LLM decides: "I need to search the web, then summarise, then draft an email"
Act (Tools)	Use tools to take actions in the world	Search the web, query a database, send an email, call an API, write a file, execute code

Example: AI Agent for Competitive Analysis

REAL-WORLD EXAMPLE — AI Agent for Competitive Analysis:

TASK: "Research our top 3 competitors and write a competitive analysis report."

SENSE: Receives task + list of competitor names.

PLAN: "I will (1) search for each competitor's latest product updates, (2) search for their pricing, (3) search for recent news, (4) synthesise findings, (5) write the report."

ACT:

- Tool call: web_search("Competitor A product updates 2024") -> results
- Tool call: web_search("Competitor A pricing") -> results
- [... 8 more tool calls ...]
- Tool call: write_file("competitive_analysis.docx", content) -> file created

RESULT: Complete competitive analysis report in 3 minutes. Human analyst: 4-6 hours.

Lesson 5.2 — Agent Planning Patterns

Rohan speaking:

There are three main planning patterns for AI agents. Choosing the right one is a critical PM decision — it affects reliability, cost, and the types of tasks your agent can handle.

Pattern	How It Works	Best For	Risk	Real Example
ReAct	Reason -> Act -> Observe -> Repeat	Research, investigation tasks	Can loop indefinitely without termination	Perplexity AI research agent

			conditions	
Plan-Execute	Create full plan first, then execute steps	Complex multi-step tasks with known structure	Plan may be wrong; hard to adapt mid-execution	Devin AI (software engineering agent)
Manager-Worker	Orchestrator agent delegates to specialist agents	Complex workflows with distinct subtasks	Coordination overhead, error propagation between agents	AutoGPT, multi-agent customer service systems

Case Study: Recruiting Agent Planning Pattern

CASE STUDY — Choosing the Right Planning Pattern for a Recruiting Agent:

SCENARIO: AI agent for a recruiting firm needs to:

(1) Search LinkedIn for candidates, (2) Screen resumes, (3) Draft outreach emails, (4) Schedule interviews.

OPTION A — ReAct: Unpredictable execution order. Might draft emails before screening resumes. REJECTED.

OPTION B — Plan-Execute: Creates plan first: "Step 1: Search. Step 2: Screen. Step 3: Draft. Step 4: Schedule." Executes in order. SELECTED.

OPTION C — Manager-Worker: Overkill for this use case. High coordination overhead. REJECTED.

RESULT: 85% task completion rate. Average time per candidate: 4 minutes vs. 45 minutes for human recruiter.

Lesson 5.3 — Tool Design for Agents

Henry speaking:

Every tool your agent uses needs a schema. The schema is how the agent knows what the tool does, when to use it, and how to call it correctly. Bad tool schemas are the #1 cause of agent failures. If the description is vague, the agent will misuse the tool.

Tool Schema Template

TOOL SCHEMA TEMPLATE:

```
{
  "name": "search_knowledge_base",
  "description": "Search the product knowledge base for relevant information. Use this when the user asks about product features, pricing, or policies. Do NOT use for general web search.",
  "parameters": {
    "query": {
      "type": "string",
      "description": "The search query. Be specific. Use keywords from the user's question."
    },
    "max_results": {
      "type": "integer",
      "description": "Maximum number of results to return. Default: 3. Max: 10."
    }
  }
}
```

```

},
"returns": "Array of (title, content, source_url) objects",
"constraints": [
  "Only search the internal knowledge base, not the web",
  "Always use at least 2 keywords in the query"
]
}

```

KEY PRINCIPLE: The description field is the most important part of the schema. It tells the agent **WHEN** to use the tool and **WHEN NOT** to use it. Vague descriptions lead to tool misuse.

Lesson 5.4 — The 5 Agent Failure Modes

Rohan speaking:

Failure Mode	What Happens	Prevention
Infinite Loop	Agent keeps calling tools without making progress	Max steps limit (e.g. 15 steps), loop detection
Tool Misuse	Agent calls the wrong tool or with wrong parameters	Clear tool descriptions, parameter validation
Context Loss	Agent forgets earlier steps in long tasks	Explicit state object, summarisation at checkpoints
Hallucinated Actions	Agent invents tool calls that do not exist	Strict tool list in system prompt, output validation
Scope Creep	Agent takes actions beyond its intended scope	Explicit action constraints, human approval for high-stakes actions

Anti-Pattern: Agent Scope Creep

CASE STUDY — Agent Failure Mode in Production:

A customer service agent at a SaaS company was given access to: search_knowledge_base, create_ticket, send_email.

FAILURE: The agent started sending emails to customers without human approval (Scope Creep).

Root cause: The send_email tool description said "Send an email to the customer." No constraint on when to use it.

FIX: Updated tool description: "Send an email to the customer **ONLY** after explicit human approval. Never send emails autonomously."

Added: Human approval gate before any send_email call.

LESSON: Every tool that takes an irreversible action needs an explicit constraint and a human approval gate.

Lesson 5.5 — Agent Reliability Checklist

Henry speaking:

Agent Reliability Checklist

TERMINATION CONDITIONS:

- ☐ Maximum steps defined (e.g. 15 steps)
- ☐ Maximum duration defined (e.g. 60 seconds)
- ☐ Success condition defined (how does the agent know it is done?)
- ☐ Failure condition defined (when should the agent give up?)

HUMAN OVERSIGHT:

- ☐ Human approval required for irreversible actions
- ☐ Human handoff path for tasks the agent cannot complete
- ☐ Audit log of all agent actions

ERROR HANDLING:

- ☐ Retry logic for transient tool failures
- ☐ Graceful degradation if a tool is unavailable
- ☐ Clear error message to user if agent fails

SAFETY:

- ☐ Agent cannot take actions outside defined scope
- ☐ PII is not logged in agent action history
- ☐ Rate limiting on expensive tool calls

Module 5 Lab Assignment

Lab Assignment

1. Choose a planning pattern for your agent and justify the choice.
Write 3 sentences explaining why you chose ReAct, Plan-Execute, or Manager-Worker.
2. Write tool schemas for all tools your agent needs.
Include: name, description, parameters, returns, constraints.
3. Define your agent's state object (full JSON schema).
What information does the agent need to track across steps?
4. Complete the Agent Reliability Checklist.
Every unchecked item is a risk. Prioritise and plan to address them.
5. Identify which of the 5 failure modes your agent is most vulnerable to.
Define your specific mitigation strategy.

PORTFOLIO TIP: This becomes Artifact 4 (Agent Architecture Document) of your AI PM Portfolio.

End of Part 1 — Modules 1 to 5

Part 1 Complete — What Is Next

You have now completed the first half of the AI Product Management Certification.

WHAT YOU HAVE LEARNED:

Module 1: How to identify high-leverage AI opportunities using the 4D Method and 5-Question Framework.

Module 2: How to design trustworthy AI experiences using the 3P Framework and 5 Trust Signals.

Module 3: How to systematically optimise AI outputs using the Prompt Engineering Hierarchy and RAG.

Module 4: How to build rigorous evaluation suites using Golden Datasets and LLM-as-Judge.

Module 5: How to architect reliable agentic AI systems using planning patterns and the Reliability Checklist.

WHAT IS COMING IN PART 2:

Module 6: Select the Perfect LLM for Cost, Latency & Quality

Module 7: Navigate AI Ethics, Safety & Responsible Deployment

Module 8: Lead AI Teams, Manage Stakeholders & Drive Adoption

Module 9: Build Your AI PM Portfolio & Land the Role

ACTION: Complete all 5 lab assignments before moving to Part 2. The labs build on each other.

AI Product Management Certification

by Product Faculty — Rohan Varma (OpenAI) & Henry Shi (Anthropic)

PART 2 — Modules 6 to 9

Expanded Lecture Notes with Real-World Case Studies

Part 2 Overview

Part 2 covers: LLM Selection & Cost Modelling | AI Ethics & Safety | Leading AI Teams & Driving Adoption
| Building Your AI PM Portfolio & Landing the Role.

MODULE 6: Select the Perfect LLM for Cost, Latency & Quality

Lesson 6.1 — The Model Selection Framework

Henry speaking:

Model selection is one of the highest-leverage decisions an AI PM makes. The wrong model costs you 10x more than necessary, or delivers 30% worse quality. Most PMs pick a model based on hype. The right approach is systematic: evaluate on three dimensions.

Dimension	Questions to Ask	Drives Toward
Quality Requirements	What is the minimum acceptable quality? What is the cost of a wrong answer?	Higher quality = larger, more capable model
Latency Requirements	What is the maximum acceptable response time? Is this real-time or async?	Lower latency = smaller, faster model
Cost Constraints	What is the budget per query? What is the expected query volume?	Lower cost = smaller model or caching strategy

Model Selection Decision Rule

MODEL SELECTION DECISION RULE:

Quality > Latency > Cost: Use Frontier model (GPT-4o, Claude 3.5 Sonnet)

Latency > Quality > Cost: Use Mid-Tier model (GPT-4o-mini, Claude 3 Haiku)

Cost > Quality > Latency: Use Small/Fast model (GPT-3.5-turbo, Claude Instant)

Privacy > everything: Use Open Source (Llama 3, Mistral) on your own infrastructure

RULE: Always start with the cheapest model that meets your quality bar. Upgrade only when you have evidence you need to.

Lesson 6.2 — The Model Tier Framework

Rohan speaking:

Tier	Models	Best For	Cost	Latency
Frontier	GPT-4o, Claude 3.5 Sonnet, Gemini 1.5 Pro	Complex reasoning, high-stakes outputs	High (\$\$\$)	Medium (2-5s)
Mid-Tier	GPT-4o-mini, Claude 3 Haiku, Gemini Flash	Most production use cases	Medium (\$)	Fast (0.5-2s)
Small/Fast	GPT-3.5-turbo, Claude Instant	Simple classification, routing, formatting	Low (\$)	Very fast (<0.5s)
Open Source	Llama 3, Mistral, Phi-3	Privacy-sensitive, on-premise, fine-tuning	Infra cost only	Depends on infra

Case Study: Klarna Model Routing

CASE STUDY — Klarna AI Model Routing:

TASK: Handle 2.3M customer service conversations/month.

INITIAL CHOICE: GPT-4 (Frontier). Monthly cost: \$180,000. Too expensive.

ANALYSIS:

70% of queries: simple (order status, refund status) — Small/Fast model sufficient

25% of queries: medium complexity (dispute resolution) — Mid-Tier sufficient

5% of queries: complex (fraud, legal, escalations) — Frontier model needed

SOLUTION: Model routing by complexity.

Simple (70%): GPT-3.5-turbo at \$0.002/query

Medium (25%): GPT-4o-mini at \$0.015/query

Complex (5%): GPT-4o at \$0.060/query

RESULT: Monthly cost: \$180,000 to \$23,000 (-87%). Quality maintained. Klarna reported \$40M annual savings.

Lesson 6.3 — The Cost Model Every AI PM Must Build

Henry speaking:

Monthly Cost Formula + Worked Example

MONTHLY COST FORMULA:

Monthly Cost =

(Avg Input Tokens per Query x Input Price per 1M tokens / 1,000,000)

+ (Avg Output Tokens per Query x Output Price per 1M tokens / 1,000,000)

x Monthly Query Volume

WORKED EXAMPLE (GPT-4o):

Input: 2,000 tokens x \$5/1M = \$0.010 per query

Output: 500 tokens x \$15/1M = \$0.0075 per query

Total per query: \$0.0175 | Monthly volume: 100,000 | Monthly cost: \$1,750

SAME EXAMPLE (GPT-4o-mini):

Input: 2,000 tokens x \$0.15/1M = \$0.0003 per query

Output: 500 tokens x \$0.60/1M = \$0.0003 per query

Total per query: \$0.0006 | Monthly cost: \$60

GPT-4o-mini is 29x cheaper. If quality is acceptable, use it.

Cost Optimisation Lever	How It Works	Typical Savings
Prompt Compression	Reduce input tokens by removing redundant context	20-30% cost reduction
Response Length Constraints	Add "Respond in maximum 150 words" to system prompt	30-50% output token reduction
Caching	Cache identical or near-identical queries	40-60% cost reduction for repetitive queries
Model Routing	Use cheap model for simple queries,	50-80% cost reduction

	expensive for complex	depending on query mix
Batching	Batch async requests instead of processing one at a time	50% cost reduction on batch-eligible workloads

Lesson 6.4 — Fine-Tuning vs. Prompt Engineering vs. RAG

Rohan speaking:

Approach	When to Use	Cost	Time to Value
Prompt Engineering	Always try first. Solves 80% of quality problems.	Low	Hours
RAG	When the model needs your specific data/knowledge	Medium	Days-Weeks
Fine-Tuning	When you need specific style, format, or domain expertise that prompting cannot achieve	High	Weeks-Months

Fine-Tuning Decision Tree + Bloomberg GPT Case Study

DECISION TREE:

Step 1: Try prompt engineering. Does quality meet your bar? YES -> Ship it. NO -> Step 2.

Step 2: Does the model lack specific knowledge/data? YES -> Add RAG. NO -> Step 3.

Step 3: Does the model need specific style/format/domain expertise? YES -> Fine-tune. NO -> Revisit your quality bar.

CASE STUDY — Bloomberg GPT:

Bloomberg fine-tuned a 50B parameter model on 700B tokens of financial data.

Result: 40% better performance on financial NLP tasks vs. general-purpose models.

This is the RIGHT use of fine-tuning: domain specialisation, not a quality shortcut.

Module 6 Lab Assignment

Lab Assignment

1. Score your AI feature on the 3-dimension matrix (Quality, Latency, Cost). Rank them 1-2-3.
2. Select a model tier and justify your choice in 3 sentences.
3. Build your cost model at 3 volume scenarios: 10K, 100K, and 1M queries/month.
4. Identify your top 2 cost optimisation levers and estimate savings.
5. Decide: Prompt Engineering, RAG, or Fine-Tuning — document your reasoning using the decision tree.

MODULE 7: Navigate AI Ethics, Safety & Responsible Deployment

Lesson 7.1 — Why Ethics Is a PM Problem, Not a Legal Problem

Henry speaking:

Henry Shi, Anthropic

"At Anthropic, we do not have a separate ethics team that reviews products. Every PM is responsible for the ethical dimensions of their product. That is the only model that works at scale. If ethics is someone else's job, it will always be deprioritised."

The Cost of Getting Ethics Wrong

THE COST OF GETTING ETHICS WRONG:

Amazon Rekognition (2018): Facial recognition misidentified 28 members of Congress as criminals. 39% error rate for darker-skinned women vs. 0% for lighter-skinned men. Product pulled from law enforcement use.

Apple Card (2019): Algorithm offered women significantly lower credit limits than men with identical financial profiles. Investigated by NY Department of Financial Services.

Air Canada Chatbot (2024): Hallucinated refund policy. Court ruled chatbot output was binding. Company lost lawsuit.

PATTERN: All three failures could have been caught with a pre-launch ethics review.

Lesson 7.2 — The 6 Ethical Dimensions Every AI PM Must Evaluate

Rohan speaking:

Dimension 1: Fairness & Bias

Test	How to Run It	Red Flag
Demographic parity	Run Golden Dataset with inputs representing different demographics	>10% quality gap between groups
Language parity	Test in all supported languages	>15% quality gap vs. English
Edge case representation	Check if Golden Dataset includes minority use cases	<10% of dataset represents edge cases

Anti-Pattern: Amazon Hiring AI

CASE STUDY — Amazon Hiring AI Bias:

Amazon built an AI resume screening tool trained on 10 years of hiring data.

Problem: 10 years of data reflected historical bias — mostly male hires in tech roles.

Result: The AI penalised resumes that included the word "women's" (e.g. "women's chess club").

Amazon scrapped the tool in 2018.

LESSON: If your training data reflects historical bias, your model will amplify it.

Dimension 2: Privacy & Data Governance

AI Privacy Checklist

AI PRIVACY CHECKLIST:

DATA COLLECTION:

- ☐ What user data does the AI receive as input?
- ☐ Is any of this data PII? (names, emails, health info, location)
- ☐ Does the user know their data is being processed by AI?
- ☐ Have users consented to AI processing of their data?

DATA STORAGE:

- ☐ Are conversation logs stored? For how long?
- ☐ Are conversation logs used for model training?
- ☐ Can users request deletion of their conversation history?

DATA TRANSMISSION:

- ☐ Is data encrypted in transit to the model provider?
- ☐ Does your model provider's DPA meet compliance requirements?
- ☐ Is any data sent to third-party tools or APIs?

Dimension 3: Transparency & Explainability

When full transparency is REQUIRED (non-negotiable):

- AI makes decisions affecting user rights (loans, hiring, insurance)
- AI generates content that could be mistaken for human-created
- AI is used in healthcare, legal, or financial contexts
- Regulations require it (EU AI Act, GDPR)

Dimension 4: Safety & Harm Prevention — 3 Lines of Defense

Line of Defense	What It Does	Examples
Line 1: Input Guardrails	Filter harmful inputs before they reach the model	Content classification, PII detection, prompt injection detection, rate limiting
Line 2: Model-Level Safety	Constrain model behaviour through system prompt	Negative instructions ("never provide..."), safe messaging guidelines for sensitive topics
Line 3: Output Guardrails	Filter harmful outputs before they reach the user	Content moderation, PII detection in outputs, factual claim flagging, human review queue

Dimension 5: Accountability Matrix

Scenario	Required Oversight Level
AI suggests a product recommendation	None — user can ignore
AI drafts an email	User reviews before sending
AI updates a database record	Human approval required
AI makes a hiring or lending decision	Human makes final decision, AI only assists
AI takes any irreversible action	Explicit human confirmation every time

Dimension 6: Reliability & Robustness

- Consistency: Same input should produce consistent quality output across runs and user segments
- Failure handling: What happens when the API is down, response is malformed, or context window is exceeded?
- Graceful degradation: Every AI feature must have a non-AI fallback path

Lesson 7.3 — The EU AI Act: PM Summary

Henry speaking:

Risk Level	AI Application Examples	Requirements
Unacceptable (Banned)	Social scoring, real-time biometric surveillance in public spaces	Do not build. Full stop.
High Risk	AI in hiring, credit scoring, medical devices, critical infrastructure	Mandatory human oversight, transparency, accuracy requirements, EU registration
Limited Risk	Chatbots, AI-generated content, deepfakes	Disclose AI interaction, label AI-generated content
Minimal Risk	Spam filters, recommendation systems, AI in video games	No specific requirements

Lesson 7.4 — The Pre-Launch Ethics Review Template

Pre-Launch Ethics Review Template

AI FEATURE ETHICS REVIEW

SECTION 1: HARM ASSESSMENT

1. What is the worst-case harm this feature could cause?
2. Probability: Very Low / Low / Medium / High
3. Severity: Minimal / Moderate / Severe / Critical
4. Risk Score = Probability x Severity

SECTION 2: FAIRNESS ASSESSMENT

5. Which user groups does this feature serve?
6. Have you tested quality across all groups? Results?
7. Are there groups who could be disproportionately harmed?

SECTION 3: TRANSPARENCY ASSESSMENT

8. Do users know they are interacting with AI? How?
9. For AI-assisted decisions: can users understand why?

10. Is there a human appeal/override mechanism?

SECTION 4: GUARDRAILS ASSESSMENT

11. What input guardrails are in place?

12. What output guardrails are in place?

13. What is the human oversight mechanism?

14. What is the incident response plan?

SECTION 5: LAUNCH DECISION

☐ APPROVED ☐ APPROVED WITH CONDITIONS ☐ DELAYED ☐ REJECTED

Signatures: PM ____ | Legal ____ | Trust & Safety ____

Module 7 Lab Assignment

Lab Assignment

1. Run a Harm Assessment — identify top 3 harms, score probability x severity for each.
2. Complete the Privacy Checklist — map every data flow in your AI feature.
3. Design your 3 Lines of Defense with specific guardrails for your use case.
4. Classify your feature under the EU AI Act. Justify your classification.
5. Run the full Pre-Launch Ethics Review template and get sign-off from one non-PM stakeholder.

MODULE 8: Lead AI Teams, Manage Stakeholders & Drive Adoption

Lesson 8.1 — The AI PM Unique Leadership Challenges

Henry speaking:

Challenge	Why It Is Different	How to Handle It
Probabilistic Outcomes	Stakeholders expect software to work 100% of the time. AI works 87% of the time.	Always anchor AI performance to a human or legacy system baseline.
The Expertise Gap	You cannot know everything about ML. But you need to ask the right questions.	Use the Informed Challenger model: "What is the failure mode?" / "Is there a simpler approach?"
Shifting Ground	Models change. GPT-4 today, GPT-5 tomorrow.	Build for model agnosticism. PRD must include: "System must switch model providers with no more than X days of engineering work."
AI Will Do Everything Expectation	Executives see ChatGPT demos and expect magic.	Set expectations before demos. Show a failure case proactively.
Evaluation Communication Gap	Technical metrics mean nothing to business stakeholders.	Use the Translation Framework: "Faithfulness: 0.87" = "13% of responses contain information not in our knowledge base."

Lesson 8.2 — Building Your AI Team

Henry speaking:

Role	Owns
AI PM (You)	Problem definition, success metrics, evaluation strategy, risk register, stakeholder communication, ethics review, roadmap
ML Engineer	Model selection, fine-tuning, evaluation infrastructure, model deployment
AI/UX Designer	Interaction patterns, trust signals, error states, user research for AI features
Data Engineer	Data pipeline, data quality, labelling, Golden Dataset infrastructure
Prompt Engineer	System prompt development, iteration, evaluation, prompt versioning

Lesson 8.3 — The AI Roadmap: Horizon Model

Rohan speaking:

Horizon	Timeframe	Criteria	Goal
NOW	0-3 months	High feasibility + High pain	Ship something real. Build credibility. Prove AI works.
NEXT	3-6 months	Medium feasibility + High impact	Build the product that drives retention and expansion.
LATER	6-12 months	Lower feasibility + Highest strategic value	Build the competitive advantage that is hard to copy.

AI Feature Prioritisation Score + Example

AI FEATURE PRIORITISATION SCORE:

Priority Score = (Business Impact x 0.30) + (Feasibility x 0.25) + (User Pain x 0.25) + (Strategic Value x 0.20)

Score each dimension 1-10. Calculate weighted score. Rank features by score.

EXAMPLE:

Feature A: Auto-fill CRM from call transcripts

Score = (9x0.30) + (8x0.25) + (9x0.25) + (7x0.20) = 2.7 + 2.0 + 2.25 + 1.4 = 8.35 -> NOW

Feature B: AI-powered sales forecasting

Score = (9x0.30) + (5x0.25) + (7x0.25) + (9x0.20) = 2.7 + 1.25 + 1.75 + 1.8 = 7.5 -> NEXT

Feature C: Autonomous deal negotiation agent

Score = (10x0.30) + (3x0.25) + (8x0.25) + (10x0.20) = 3.0 + 0.75 + 2.0 + 2.0 = 7.75 -> LATER

Lesson 8.4 — The 4 Stakeholder Archetypes

Henry speaking:

Archetype	Profile	How to Win Them
The Skeptic	"AI is just hype. Show me the ROI."	Lead with baseline comparison. Hard numbers. Working prototype with real metrics.
The Enthusiast	"AI can do everything!"	Channel their energy. "Here is what we can achieve in 90 days, and here is the roadmap to your bigger vision."
The Risk Manager	"What if it goes wrong?"	Show your risk register. "Here are the 5 failure modes and our mitigation for each."
The End User Champion	"Will this actually help users?"	Involve them in user research. Show real user feedback from pilots.

Lesson 8.5 — The 5 Barriers to AI Adoption

Rohan speaking:

Barrier	Root Cause	How to Overcome It
Trust Deficit	Users have been burned by AI errors before	Show sources, show confidence levels, start with low-stakes use cases
Workflow Disruption	AI requires users to change how they work	Meet users where they are, make AI opt-in, show time savings explicitly
Black Box Problem	Users do not understand why AI made a recommendation	Explain reasoning, show inputs, make uncertainty visible
Fear of Replacement	Users worry AI will take their job	Frame as augmentation, involve users in design, show how AI makes them look better
Learning Curve	Users do not know how to get good outputs from AI	Provide prompt templates, show examples of great outputs, build progressive disclosure

Lesson 8.6 — AI Adoption Metrics

Henry speaking:

Metric	What It Measures	Target	Red Flag
AI Acceptance Rate	% of AI suggestions users accept without editing	>60%	<40% = trust or quality problem
AI Override Rate	% of AI outputs users significantly edit or reject	<30%	>50% = quality problem
AI Return Rate	% of users who use AI feature again after first use	>50%	<30% = value problem
Time-to-Value	Time from first AI use to first "aha moment"	<5 minutes	>15 minutes = onboarding problem

Adoption Health Score + Slack Case Study

ADOPTION HEALTH SCORE:
Score = (Acceptance Rate x 0.30) + (Return Rate x 0.25) + (Task Completion Rate x 0.25) + (User Satisfaction x 0.20)
Green: > 75 | Yellow: 50-75 | Red: < 50

CASE STUDY — Slack AI Adoption:
Slack launched AI channel summaries in 2024. Initial Adoption Health Score: 42 (Red).
Root cause: Summaries were too long and missed key action items.
Fixes: Reduced summary length from 200 to 80 words. Added "Action Items" section. Added feedback loop.
Result after 6 weeks: Adoption Health Score: 71 (Yellow, approaching Green).

Lesson 8.7 — The 5-Phase AI Launch Playbook

Rohan speaking:

Phase	Timing	Key Actions	Success Criteria
Seed	Weeks 1-2	10-20 power users, early access, qualitative feedback, case studies	3+ positive case studies, no critical bugs
Expand	Weeks 3-4	10% rollout, daily metric monitoring, weekly office hours	Adoption Health Score > 50
Educate	Weeks 5-6	Getting Started guide, 3-min demo video, Tips & Tricks email	Feature discovery rate > 40%
Normalize	Weeks 7-8	Full rollout, AI becomes default, celebrate adoption milestones	Adoption Health Score > 65
Optimize	Ongoing	Monthly adoption review, quarterly user interviews, continuous prompt iteration	Adoption Health Score > 75

Module 8 Lab Assignment

Lab Assignment

- 1. Map your stakeholders using the 4 archetypes — write a 3-sentence communication plan for each.
- 2. Build your AI Roadmap using the Horizon Model — score 5 candidate features using the Priority Score formula.
- 3. Design your Adoption Strategy — identify top 2 barriers and mitigation plans.

4. Build your Stakeholder Communication Pack (weekly, monthly, quarterly templates).
5. Define your Adoption Health Score with targets for all 4 metrics.

MODULE 9: Build Your AI PM Portfolio & Land the Role

Lesson 9.1 — The AI PM Job Market Reality

Henry speaking:

The AI PM Opportunity

THE OPPORTUNITY:

Demand for AI PMs is growing 40% year-over-year.
Most PMs have surface-level AI knowledge ("I use ChatGPT").
Genuinely qualified AI PMs who can build eval suites, write AI PRDs, and architect agents are scarce.

If you have completed the labs in this course, you are already in the top 10% of AI PM candidates.

Profile	Description	Key Differentiator
The Builder	Has shipped AI products. Has a portfolio. Can talk about failures as fluently as successes.	Commands a premium. Rare.
The Translator	Deep domain expertise (healthcare, finance, legal) + AI knowledge.	Identifies AI opportunities generalists miss.
The Evaluator	Obsessed with quality and measurement. Has built eval suites. Understands LLM-as-Judge.	Rare and highly valued. Most PMs cannot do this.

Lesson 9.2 — The 5-Artifact AI PM Portfolio

Rohan speaking:

The 5 Portfolio Artifacts

ARTIFACT 1: AI Opportunity Assessment (2 pages)

Demonstrates: Problem identification, business thinking, AI suitability judgment.

Must include: Pain Score, AI Suitability Score, Business Case, Risk Register.

THIS IS: Module 1 Lab Assignment.

ARTIFACT 2: The AI PRD (4-6 pages)

Demonstrates: Requirements writing, AI-specific thinking, cross-functional communication.

Must include all AI-specific sections: AI Approach, Data Requirements, Model Selection, Prompt Strategy, Evaluation Plan, Guardrails, Cost Model, Risk Register.

ARTIFACT 3: The Eval Suite

Demonstrates: Quality obsession, technical depth, measurement discipline.

Must include: 30-example Golden Dataset, custom judge prompt, scoring spreadsheet, failure analysis, improvement log.

THIS IS: Module 4 Lab Assignment.

ARTIFACT 4: Agent Architecture Document

Demonstrates: Systems thinking, agentic AI understanding, technical communication.

Must include: Agent Overview, Agent Stack, Tool Schemas, State Object, Termination Conditions, Failure Mode Analysis, Reliability Checklist.

THIS IS: Module 5 Lab Assignment.

ARTIFACT 5: STAR-AI Case Study

Demonstrates: Storytelling, impact measurement, learning mindset.

Must include: Situation, Task, Action (AI-specific), Result + eval scores before/after, cost model at scale.

STAR-AI Framework (Detailed)

STAR-AI Framework

SITUATION: Business context, problem, baseline (before AI). What was the pain? What was the cost?

TASK: Your specific role, responsibilities, constraints. What were you asked to do?

ACTION (AI-Specific):

- AI approach chosen and why (RAG, Agent, Classifier, Generator)
- Evaluation criteria defined upfront
- Prompt iterations (v1 to v2 to v3 with quality scores)
- Failure modes encountered and how you handled them
- Responsible AI considerations and guardrails built

RESULT:

- Key metric change (before vs. after)
- Cost/latency impact
- Learnings: what worked, what did not, what you would do differently

AI ADDITIONS:

- Eval scores before/after (Faithfulness, Relevance, Completeness)
- Cost model at scale (monthly cost at 100K queries)
- Failure rate and how you reduced it
- Next steps: what would you build next?

Lesson 9.3 — The 3-Platform Portfolio Strategy

Rohan speaking:

Platform	What to Publish	Goal	Frequency
LinkedIn	1 post per artifact (5 total), learnings not just outputs, #AIProductManagement	Visibility and inbound opportunities	1 post per week for 5 weeks
GitHub	Eval suite code, Golden Dataset (anonymised), prompt templates	Technical credibility signal	Publish once, keep updated
Personal Site/Notion	All 5 artifacts linked, AI PM Philosophy statement, contact info	Single URL to share with recruiters	Build once, update quarterly

Lesson 9.4 — The AI PM Interview Playbook

Henry speaking:

Interview Type 1: AI Opportunity Question

TYPE 1: THE AI OPPORTUNITY QUESTION

"How would you use AI to improve [product]?"

STRUCTURE (2.5 minutes):

1. Identify highest-pain user workflow (30 sec)
2. Apply AI Suitability Test (30 sec)
3. Define specific AI approach — RAG, Agent, Classifier, or Generator (30 sec)
4. Define success metrics (30 sec)
5. Identify top risk (30 sec)

Interview Type 2: Technical Depth Question

TYPE 2: THE TECHNICAL DEPTH QUESTION

"Walk me through how RAG works."

STRUCTURE:

1. Plain English explanation (what problem does RAG solve?)
2. When to use it vs. alternatives
3. The 4 quality levers (chunking, retrieval, context window, grounding)
4. When RAG fails
5. Specific example from your portfolio

Interview Type 3: Metrics Question

TYPE 3: THE METRICS QUESTION

"How would you measure success of an AI feature?"

USE THE 4-LAYER KPI FRAMEWORK:

Layer 1 — Model: Faithfulness, Relevance, Task Completion Rate

Layer 2 — Agent (if applicable): Tool call success rate, Step completion rate

Layer 3 — Product: Feature adoption rate, AI Acceptance Rate, AI Override Rate, Time-to-Value

Layer 4 — Business: Revenue impact, Cost savings, NPS delta, Retention impact

Interview Type 4: Failure Question

TYPE 4: THE FAILURE QUESTION

"Tell me about an AI project that did not go as planned."

STRUCTURE:

1. What you built and what you expected (30 sec)
2. What actually happened — be specific with numbers (30 sec)
3. Root cause diagnosis (45 sec)
4. What you changed — specific actions taken (30 sec)
5. What you would do differently from day one (30 sec)

Interview Type 5: Responsible AI Question

TYPE 5: THE RESPONSIBLE AI QUESTION

"How do you approach AI ethics in your products?"

STRUCTURE:

1. Ethics as product quality, not compliance (15 sec)
2. The 6 dimensions you evaluate (30 sec)
3. Your pre-launch ethics review process (30 sec)
4. A specific ethical trade-off you navigated (45 sec)
5. Your personal philosophy (15 sec)

Lesson 9.5 — AI PM Salary Landscape (2024-2025)

Rohan speaking:

Level	Company Type	Base Salary	Total Comp
PM II (Mid)	Big Tech (Google, Meta, Microsoft)	\$160K-\$200K	\$250K-\$350K
PM II (Mid)	AI Startup (Series B+)	\$140K-\$180K	\$200K-\$400K (with equity)
Senior PM	Big Tech	\$200K-\$250K	\$350K-\$550K
Senior PM	AI Startup	\$170K-\$220K	\$300K-\$700K (with equity)
Principal/Staff PM	Big Tech	\$250K-\$320K	\$500K-\$900K

How to Command Top of Range

HOW TO COMMAND THE TOP OF THE RANGE:

1. Have a public portfolio with all 5 artifacts. Most candidates have zero.
2. Be able to demo a working AI product you built. Not a slide deck — a working product.
3. Have eval suite experience: "I built a 200-example Golden Dataset and reduced hallucination rate from 18% to 4%."
4. Have a specific domain expertise + AI combination: Healthcare AI PM, Legal AI PM, Fintech AI PM.
5. Have shipped something real. Even a side project. Even a prototype. Builders get hired.

Module 9 Lab Assignment — Your Final Portfolio Sprint

Final Portfolio Sprint

WEEK 1: Assemble all 5 artifacts from your lab assignments.

Artifact 1: AI Opportunity Assessment (Module 1 Lab)

Artifact 2: AI PRD extended with all AI-specific sections (Modules 1, 3, 4, 6, 7)

Artifact 3: Eval Suite (Module 4 Lab)

Artifact 4: Agent Architecture Document (Module 5 Lab)

Artifact 5: STAR-AI Case Study (write fresh using your best lab project)

WEEK 2: Publish your portfolio.

Day 1: Create your Notion/personal site with all 5 artifacts linked.

Day 2: Publish Artifact 1 on LinkedIn with 3 key learnings.

Day 3: Publish your Eval Suite on GitHub.

Day 4: Publish Artifact 5 (STAR-AI Case Study) on LinkedIn.

Day 5: Write and publish your AI PM Philosophy Statement.

WEEK 3: Start applying.

Target: 3 AI PM roles per week.

Personalise your cover letter with 1 specific AI opportunity for each company.

Share your portfolio URL in every application.

Course Complete — Full Checklist

Course Complete

WHAT YOU HAVE BUILT ACROSS ALL 9 MODULES:

Module 1: AI Opportunity Assessment + 1-page AI PRD

Module 2: Interaction Pattern Map + 3P Framework Score + Trust Signal Design + Error State Design

Module 3: Production System Prompt + Prompt Hierarchy Test + RAG Architecture Design

Module 4: 30-example Golden Dataset + LLM-as-Judge Prompt + Failure Analysis + Improvement Log

Module 5: Agent Planning Pattern Decision + Tool Schemas + State Object + Reliability Checklist

Module 6: Model Selection Decision + Cost Model (3 scenarios) + Cost Optimisation Plan

Module 7: Harm Assessment + Privacy Checklist + 3 Lines of Defense + Ethics Review

Module 8: Stakeholder Map + AI Roadmap + Adoption Strategy + Communication Pack

Module 9: 5-Artifact Portfolio + LinkedIn Posts + GitHub Repo + Personal Site

YOU ARE NOW AN AI PM.

The frameworks are yours. The templates are yours. The portfolio is yours.

Go build something real. Ship it. Measure it. Improve it.

That is what AI PMs do.